

- Good thermal conductivity – conducts heat away

7(a)

the value of the direct current
that dissipates thermal energy at the same rate in a resistor

allow 'same power' and 'same heating effect'

7(b)

power $\propto I^2$ or $P = I^2 R$ or $P = IV$

ratio = 2.0 (allow 1 s.f.)

7(c)(i)

6.0 A (allow 1 s.f.)

Qns

7(c)(ii)

$$\sqrt{\frac{2^2 + 6^2 + 0^2}{3}}$$

$$= 3.7 \text{ A}$$

8(a)(i)

A, highest output at 10 m s^{-1}